

- 5 (a) (i) energy converted from chemical to electrical when charge flows through cell or round complete circuit B1
- (ii) (resistance of the cell) causing loss of voltage or energy loss in cell B1 [2]
- (b) (i) $E_B - E_A = I(R + r_B + r_A)$
 $12 - 3 = I(3.3 + 0.1 + 0.2)$
 $I = 2.5 \text{ A}$ C1
A1 [2]
- (ii) Power = $E \times I$
 $= 12 \times 2.5$
 $= 30 \text{ W}$ C1
A1 [2]
- (iii) $P = I^2 \times R$ or $P = V^2 / R$ or $P = VI$
 $= (2.5)^2 \times 3$ $= 9^2 / 3.6$ $= 9 \times 2.5$ C1
 $= 22.5 \text{ J s}^{-1}$ A1 [2]
- (c) power supplied from cell B is greater than energy lost per second in circuit B1 [1]
- 6 (a) (i) to produce coherent sources or constant phase difference B1 [1]